GLOBAL ACADEMY OF TECHNOLOGY

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Ideal Homes Township, Rajarajeshwari Nagar, Bengaluru-98

Department of Information Science and Engineering

Laboratory Manual

BIG DATA ANALYTICS (ISE23701) (Integrated)

Semester: 7th
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Vision of the Institute

Become a premier institution imparting quality education in engineering and management to meet the changing needs of society.

Mission of the Institute

M1: Create environment conductive for continuous learning through quality teaching and learning processes supported by modern infrastructure.

M2: Promote research and innovation through collaboration with industries.

M3: Inculcate ethical values and environmental conscious through holistic education programs.

Program Outcomes (POs)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Syllabus

Subject Code	ISE23701	CIE Marks	50
Hours/Week (L: T: P)	3:0:2	SEE Marks	50
Total Hours	50	Examination Hours	03
No. of Credits: 04			

Course Objectives: The course will enable students to:

CLO1	Understand the fundamentals of Big Data Analytics
CLO2	Explore the Hadoop framework and Hadoop Distributed File system
CLO3	Illustrate the concepts of NoSQL using MongoDB and Cassandra for Big Data
CLO4	Employ MapReduce programming model to process the big data
CL05	Understand various machine learning algorithms for Big Data Analytics, Web Mining and Social Network Analysis.

Pg. No.	PROGRAMMING EXERCISES
1.	Write a Python program to load a CSV dataset using the Pandas library and display the dataset information such as shape, column names, data types, and summary statistics.
2.	Write a Python program to preprocess a CSV dataset by performing the following operations: • Handle missing values • Remove duplicate records Rename column names
3.	Write a Python program to simulate HDFS file operations using the os and shutil libraries by creating folders, copying files, moving files, and deleting files.
4.	Write a Python program to analyze a large CSV dataset using Pandas by performing: • Filtering • Sorting • Group By Aggregation
5.	Write a Python program to store employee/student records in JSON format and perform CRUD (Create, Read, Update, Delete) operations.
6.	Write a Python program to perform document search and filtering on JSON data to simulate MongoDB query operations.
7.	Write a Python program to implement the Word Count algorithm using Python dictionaries based on the MapReduce programming approach.

8.	Write a Python program to find the Maximum, Minimum, Average, and Sum of a list of numbers using the MapReduce programming approach in Python.
9.	Write a Python program to implement Linear Regression using Scikit-Learn on a simple dataset and display the regression line and predicted values.
10.	Write a Python program to implement K-Means Clustering on the Iris dataset using Scikit-Learn and display the clustering results.

Course Outcomes

Upon successful completion of this course, students will be able to

ISE23701.1	Understand the fundamentals of Big Data Analytics and its applications.
ISE23701.2	Describe Hadoop and its ecosystem and Hadoop Distributed File system.
ISE23701.3	Demonstrate the concepts of NoSQL using MongoDB and Cassandra Databases for Big Data.
ISE23701.4	Illustrate the MapReduce programming model to process the big data along with Hadoop tools.
ISE23701.5	Apply the knowledge of Hadoop framework, HDFS, MongoDB, Casandra, Machine learning algorithms in solving the real-world problems associated with Big Data.

CO-PO Mapping

CO/PO Mapping																
CO/PO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**	**PSO4**
ISE23701.1	3	2	2	2	1	1	-	1	-	-	-	2	-	3	-	-
ISE23701.2	3	2	2	2	1	1	-	1	-	-	-	2	1	3	-	-
ISE23701.3	3	2	2	2	1	1	-	1	-	-	-	2	1		-	3
ISE23701.4	3	2	2	2	1	1	-	1	-	-	-	2	-	3	-	-
ISE23701.5	3	2	2	2	1	1	-	1	-	-	-	2	-	3	-	1
Average	3	2	2	2	1	1	-	1	-	-	-	2	1	3	-	2

Scheme of Examination:

Assessment Details:

- The weightage of Continuous Lab Assessment is 50% and lab internal marks is 50% and total is 20. Out of total 20 marks the minimum passing marks is 8.
- Each experiment is to be evaluated for conduction with an **observation sheet and record write-up**.
- Rubrics for the evaluation of the journal/write-up for hardware/software experiments are designed by the faculty who is handling the laboratory session.
- In a test, test write-up, conduction of experiment, acceptable result, and procedural knowledge will carry a weightage of 60% and the rest 40% for viva-voce.
- The suitable rubrics can be designed to evaluate each student's performance and learning ability

Table: Distribution of weightage for CIE

	Component	Marks	Total Marks
CIE	Observation Book	**5**	**20**
	Record Book	**5**	
	Internal Assessment Test	**10**	
Grand Total			**20**

Marks Distribution

Component	Max Marks	Marks Distribution	Total Marks
Observation Book	5	Write-up - 4 Marks Execution - 12 Marks Viva Voce - 04 Marks (Round-off to 5)	**10**
Record Book	5	Write-up and on time submission - 10 Marks (Round-off to 5)	
Internal Assessment Test (Conducted for **50** and reduced to **10)**	10	Write-up - 10 Marks Execution - 30 Marks Viva Voce - 10 Marks	**10**
Total Marks			**20**

Text Books:

1. Raj Kamal and PreetiSaxena, "Big Data Analytics Introduction to Hadoop, Spark, and Machine-Learning", McGraw Hill Education, 2018 ISBN: 9789353164966, 9353164966
2. Douglas Eadline, "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem", 1 stEdition, Pearson Education, 2016. ISBN13: 978-9332570351

Reference Books:

1. Tom White, "Hadoop: The Definitive Guide", 4th Edition, O"Reilly Media, 2015.ISBN-13: 978- 9352130672 2.
2. Boris Lublinsky, Kevin T Smith, Alexey Yakubovich, "Professional Hadoop Solutions", 1 stEdition, Wrox Press, 2014ISBN-13: 978-8126551071
3. Eric Sammer, "Hadoop Operations: A Guide for Developers and Administrators",1 stEdition, O'Reilly Media, 2012.ISBN-13: 978-9350239261
4. ArshdeepBahga, Vijay Madisetti, "Big Data Analytics: A Hands-On Approach", 1st Edition, VPT Publications, 2018. ISBN-13: 978-0996025577

Web Reference:

1 http://nptel.ac.in/
2 https://www.khanacademy.org/
3 E-learning: www.vtu.ac.in
4 https://onlinecourses.nptel.ac.in/noc20_cs92/preview
5 https://www.classcentral.com/course/big-data-analytics-18252

Lab Rubrics

Rubrics for Evaluation of Observation Book

Attribute (Max. Marks)	Excellent 04	Very Good 03	Good 02	Satisfactory 01
Write-up (04)	Able to write Complete program without errors Make note of Input and expected output for all test cases.	Able to write Complete program without errors Make note of Input and expected output for only few test cases.	Able to write Complete program without errors Unable to make note of Input and expected output	Unable to write Complete program without errors Unable to make note of Input and expected output for all test cases
Attribute (Max. Marks)	**Excellent 10-12**	**Very Good 07-09**	**Good 04-06**	**Satisfactory 01-03**
Execution (12)	Able to Debugs the program independently. Able to Executed the program for all possible inputs Able to analyze the output properly	Able to Debugs the program independently. Able to Executed the program for all possible inputs Unable to analyze the output properly	Able to Debugs the program independently. Unble to Executed the program for all possible inputs Unable to analyze the output properly	Unable to Debugs the program independently. Unable to Executed the program for all possible inputs Unable to analyze the output properly
Attribute (Max. Marks)	**Excellent 04**	**Very Good 03**	**Good 02**	**Satisfactory 01**
Viva Voce (04)	Able to explain the logic of the program. Able to Answer all questions.	Able to explain the logic of the program. Able to Answer 50% of questions.	Able to explain the logic of the program. Unable to Answer any questions	Unable to explain the logic of the program. Unable to Answer any questions

Rubrics for Evaluation of Record Book

Attribute (Max. Marks)	Excellent 08-10	Very Good 05-07	Good 03-04	Satisfactory 01-02
Presentation and Periodic submission (10)	Writing the program without errors, all possible input/output are documented, completed index, and submitted on time	Writing the program without errors, only few input/outputs are documented, completed index, and submitted on time	Writing the program with errors, only few input/output are documented, incomplete index, and submitted on time.	Writing the program with errors, only few input/output are documented, incomplete index, and not submitted on time.

Rubrics for Evaluation of Internal Assessment Test

Attribute (Max. Marks)	Excellent 08-10	Very Good 05-07	Good 03-04	Satisfactory 01-02
Write-up (10)	Able to write Complete program without errors Make note of Input and expected output for all test cases.	Able to write Complete program without errors Make note of Input and expected output for only few test cases.	Able to write Complete program without errors Unable to make note of Input and expected output	Unable to write Complete program without errors Unable to make note of Input and expected output for all test cases
Attribute (Max. Marks)	**Excellent 22-30**	**Very Good 16--22**	**Good 07-15**	**Satisfactory 01-06**
Execution (30)	Able to Debugs the program independently. Able to Executed the program for all possible inputs Able to analyze the output properly	Able to Debugs the program independently. Able to Executed the program for all possible inputs Unable to analyze the output properly	Able to Debugs the program independently. Unable to Executed the program for all possible inputs Unable to analyze the output properly	Unable to Debugs the program independently. Unable to Executed the program for all possible inputs Unable to analyze the output properly
Attribute (Max. Marks)	**Excellent 08-10**	**Very Good 05-07**	**Good 03-04**	**Satisfactory 01-02**
Viva Voce (10)	Able to explain the logic of the program. Able to Answer all questions.	Able to explain the logic of the program. Able to Answer 50% of questions.	Able to explain the logic of the program. Unable to Answer any questions	Unable to explain the logic of the program. Unable to Answer any questions

Course Outcomes mapping with Bloom's Taxonomy

Sl. No.	Program	Mapped CO	Justification	Bloom's Level
1	Write a Python program to load a CSV dataset using the Pandas library and display the dataset information such as shape, column names, data types, and summary statistics.	CO2	Involves analyzing behavior due to changes in network parameters (queue/bandwidth) in protocol layers.	Analyze *(Level 4)*
2	Write a Python program to preprocess a CSV dataset by handling missing values, removing duplicate records, and renaming column	CO2	Applies data preprocessing techniques to clean and prepare	Apply *(Level 3)*

	names.		datasets for analysis.	
3	Write a Python program to simulate HDFS file operations using the os and shutil libraries by creating folders, copying files, moving files, and deleting files	CO2	Demonstrates file management operations similar to HDFS using Python libraries.	Apply (Level 3)
4	Write a Python program to analyze a large CSV dataset using Pandas by performing filtering, sorting, group by, and aggregation.	CO3	Involves analyzing datasets using data manipulation and aggregation techniques.	Analyze (Level 4)
5	Write a Python program to store employee/student records in JSON format and perform CRUD (Create, Read, Update, Delete) operations.	CO2	Applies JSON data storage and CRUD operations for structured data management.	Apply (Level 3)
6	Write a Python program to perform document search and filtering on JSON data to simulate MongoDB query operations.	CO3	Analyzes JSON documents by applying search and filtering operations similar to MongoDB queries.	Analyze (Level 4)
7	Write a Python program to implement the Word Count algorithm using Python dictionaries based on the MapReduce programming approach.	CO2	Applies the MapReduce programming model to process text data using Python dictionaries	Apply (Level 3)
8	Write a Python program to find the Maximum, Minimum, Average, and Sum of a list of numbers using the MapReduce programming approach in Python.	CO2	Applies the MapReduce approach to perform numerical data processing and aggregation.	Apply (Level 3)
9	Write a Python program to implement Linear Regression using Scikit-Learn on a simple dataset and display the regression line and predicted values.	CO3	Analyzes data by implementing a supervised machine learning algorithm for prediction.	Analyze (Level 4)
10	Write a Python program to implement K-Means Clustering on the Iris dataset using Scikit-Learn and display the clustering results.	CO3	Analyzes data by applying an unsupervised machine learning algorithm to identify clusters.	Analyze (Level 4)



Fig Bloom's Taxonomy

CHAPTER 1

INTRODUCTION

Big Data Analytics is the process of collecting, processing, analyzing, and interpreting extremely large and complex datasets to discover meaningful patterns, trends, and insights that support informed decision-making. Traditional data processing tools are often unable to handle the volume, velocity, and variety of big data, making advanced analytics techniques and distributed computing frameworks essential.

Big data is generated from multiple sources such as social media, online transactions, sensors, mobile devices, healthcare systems, and Internet of Things (IoT) devices. Organizations use Big Data Analytics to improve operational efficiency, predict future trends, enhance customer experiences, detect fraud, and optimize business processes.

Big Data Analytics combines technologies such as Hadoop, Spark, NoSQL databases, MapReduce, Machine Learning, and Data Mining to process structured, semi-structured, and unstructured data efficiently. It enables organizations to transform raw data into actionable insights that drive innovation and competitive advantage.

Characteristics of Big Data (5 Vs)

- Volume: Massive amounts of data generated every second.
- Velocity: High speed at which data is created, processed, and analyzed.
- Variety: Different types of data, including structured, semi-structured, and unstructured data.
- Veracity: Ensuring the accuracy and reliability of data.
- Value: Extracting meaningful insights that benefit organizations.

Applications of Big Data Analytics

- Healthcare (disease prediction and patient monitoring)
- Banking and Finance (fraud detection and risk analysis)
- E-commerce (recommendation systems and customer behavior analysis)
- Social Media (sentiment analysis and trend detection)
- Telecommunications (network optimization and predictive maintenance)
- Manufacturing (quality control and predictive maintenance)

Benefits of Big Data Analytics

- Improves decision-making through data-driven insights.
- Identifies hidden patterns and trends.
- Enhances operational efficiency and productivity.
- Supports predictive and prescriptive analytics.

CHAPTER 2

About the Google Colab, Programming language and Basic commands of python

About Google Colab

Google Colaboratory, commonly known as **Google Colab**, is a free cloud-based platform developed by Google that enables users to write, execute, and share Python programs directly through a web browser without installing any software. It is widely used in data science, machine learning, artificial intelligence, and big data analytics because it provides a ready-to-use programming environment with many pre-installed libraries such as Pandas, NumPy, Matplotlib, and Scikit-Learn. Google Colab also offers free access to GPUs and TPUs for computationally intensive tasks, making it suitable for developing and testing machine learning models. Additionally, it integrates seamlessly with Google Drive, allowing users to store datasets, notebooks, and project files securely while supporting real-time collaboration similar to Google Docs.

About Python Programming Language

Python is a high-level, interpreted, and object-oriented programming language developed by **Guido van Rossum** and released in **1991**. It is one of the most popular programming languages due to its simple syntax, readability, and extensive library support. Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming, making it suitable for a wide range of applications. It is extensively used in data science, big data analytics, artificial intelligence, machine learning, web development, automation, scientific computing, and cloud computing. Python's platform independence, open-source nature, and vast community support have made it the preferred language for beginners as well as professional developers.

Basic Commands of Python

Python provides a variety of basic commands that form the foundation of programming. The print() function is used to display output on the screen, while the input() function accepts data from the user during program execution. Variables are used to store data values and do not require explicit declaration of data types. Python supports built-in data types such as integers, floating-point numbers, strings, lists, tuples, dictionaries, and Boolean values. Arithmetic operators (+, -, *, /, %) are used to perform mathematical operations, whereas comparison operators (==, !=, >, <) are used to compare values. Decision-making statements such as if, if-else, and elif control the flow of execution based on conditions. Looping statements
such as for and while are used to execute a block of code repeatedly. Functions are defined using the def

keyword to create reusable code blocks. Python also allows importing external libraries using the import statement, enabling users to perform advanced tasks such as data analysis with Pandas, numerical computation with NumPy, and machine learning using Scikit-Learn. These fundamental commands provide the building blocks for developing applications ranging from simple scripts to complex data analytics solutions.

CHAPTER 3

LABORATORY PROGRAMS

1. Write a Python program to load a CSV dataset using the Pandas library and display the dataset information such as shape, column names, data types, and summary statistics.

```
# Import pandas
import pandas as pd

# Load the dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Display first 5 rows
print(data.head())

# Display number of rows and columns
print("\nRows and Columns:", data.shape)
```

Output

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa

Rows and Columns: (150, 5)
```

2. Write a Python program to preprocess a CSV dataset by performing the following operations:

   - Handle missing values
   - Remove duplicate records
   - Rename column names

```
# Import pandas
import pandas as pd

# Load the dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Handle missing values
data = data.fillna(0)

# Remove duplicate records
data = data.drop_duplicates()

# Rename column names
data = data.rename(columns={
    "sepal_length": "SepalLength",
    "sepal_width": "SepalWidth",
    "petal_length": "PetalLength",
    "petal_width": "PetalWidth",
    "species": "Species"
})

# Display the first 5 rows
 print(data.head())
```

Output :

	SepalLength	SepalWidth	PetalLength	PetalWidth	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

3. Write a Python program to simulate HDFS file operations using the os and shutil libraries by creating folders, copying files, moving files, and deleting files.

```
import os
import shutil

# Create a folder
os.mkdir("HDFS")

# Create a sample file
with open("sample.txt", "w") as f:
    f.write("Welcome to Big Data Analytics")

# Copy file to HDFS folder
shutil.copy("sample.txt", "HDFS")

# Move file
shutil.move("sample.txt", "HDFS/sample.txt")

# Display files in HDFS folder
print("Files in HDFS Folder:")
print(os.listdir("HDFS"))

# Delete the file
os.remove("HDFS/sample.txt")

print("File Deleted Successfully")
```

Output:

```
Files in HDFS Folder:
['sample.txt']

File Deleted Successfully
```

4. Write a Python program to analyze a large CSV dataset using Pandas by performing: Filtering
   Sorting
   Group By Aggregation

```
import pandas as pd

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Filtering
print("Filtered Data:")
print(data[data["species"] == "setosa"].head())

# Sorting
print("\nSorted Data:")
print(data.sort_values("sepal_length").head())

# Group By
print("\nGroup By:")
print(data.groupby("species").size())

# Aggregation
print("\nAverage Sepal Length:")
print(data.groupby("species")["sepal_length"].mean())
```

Output :

```
Filtered Data:
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
...

Sorted Data:
   sepal_length  sepal_width  petal_length  petal_width species
13           4.3          3.0           1.1          0.1  setosa
...

Group By:
species
setosa        50
versicolor    50
virginica     50

Average Sepal Length:
species
setosa        5.006
versicolor    5.936
virginica     6.588
Name: sepal_length, dtype: float64
```

5. Write a Python program to store employee/student records in JSON format and perform CRUD (Create, Read, Update, Delete) operations.

```
import json

# Create
employees = [
    {"id": 1, "name": "Ram", "salary": 25000},
    {"id": 2, "name": "Ravi", "salary": 30000}
]

# Read
print("Employee Records:")
print(json.dumps(employees, indent=4))

# Update
employees[0]["salary"] = 28000

# Delete
employees.pop(1)

print("\nAfter Update and Delete:")
print(json.dumps(employees, indent=4))
```

Output :

```
Employee Records:

[
    {
        "id": 1,
        "name": "Ram",
        "salary": 25000
    },
    {
        "id": 2,
        "name": "Ravi",
        "salary": 30000
    }
]

After Update and Delete:

[
    {
        "id": 1,
        "name": "Ram",
        "salary": 28000
    }
]
```

6. Write a Python program to perform document search and filtering on JSON data to simulate MongoDB query operations.

```
import json

students = [
    {"id": 1, "name": "Rahul", "course": "CSE"},
    {"id": 2, "name": "Anu", "course": "ISE"},
    {"id": 3, "name": "Kiran", "course": "CSE"}
]

# Search
print("Students in CSE")

for student in students:
    if student["course"] == "CSE":
        print(student)
```

Output :

```
Students in CSE

{'id': 1, 'name': 'Rahul', 'course': 'CSE'}
{'id': 3, 'name': 'Kiran', 'course': 'CSE'}
```

7. Write a Python program to implement the Word Count algorithm using Python dictionaries based on the MapReduce programming approach.

```
text = "big data analytics big data"

words = text.split()

word_count = {}

for word in words:
    if word in word_count:
        word_count[word] += 1
    else:
        word_count[word] = 1

print(word_count)
```

Output :

```
{'big': 2, 'data': 2, 'analytics': 1}
```

8. Write a Python program to find the Maximum, Minimum, Average, and Sum of a list of numbers using the MapReduce programming approach in Python.

```
numbers = [10, 20, 30, 40, 50]

print("Maximum:", max(numbers))
print("Minimum:", min(numbers))
print("Sum:", sum(numbers))
print("Average:", sum(numbers)/len(numbers))
```

Output :

```
Maximum: 50
Minimum: 10
Sum: 150
Average: 30.0
```

9. Write a Python program to implement Linear Regression using Scikit-Learn on a simple dataset and display the regression line and predicted values.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

X = data[["petal_length"]]
y = data["sepal_length"]

model = LinearRegression()
model.fit(X, y)

prediction = model.predict(X)

print("Predicted Values:")
print(prediction[:5])

plt.scatter(X, y)
plt.plot(X, prediction, color="red")
plt.show()
```

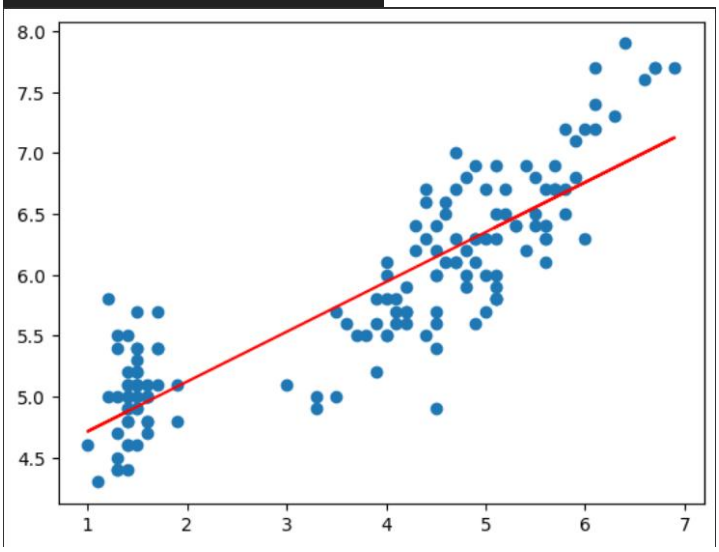
Dataset Link

- https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv

Output :

```
Predicted Values:

[4.705 4.705 4.633 4.777 4.705]
```



10. Write a Python program to implement K-Means Clustering on the Iris dataset using Scikit-Learn and display the clustering results.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

X = data[["sepal_length", "sepal_width"]]

kmeans = KMeans(n_clusters=3, random_state=0)

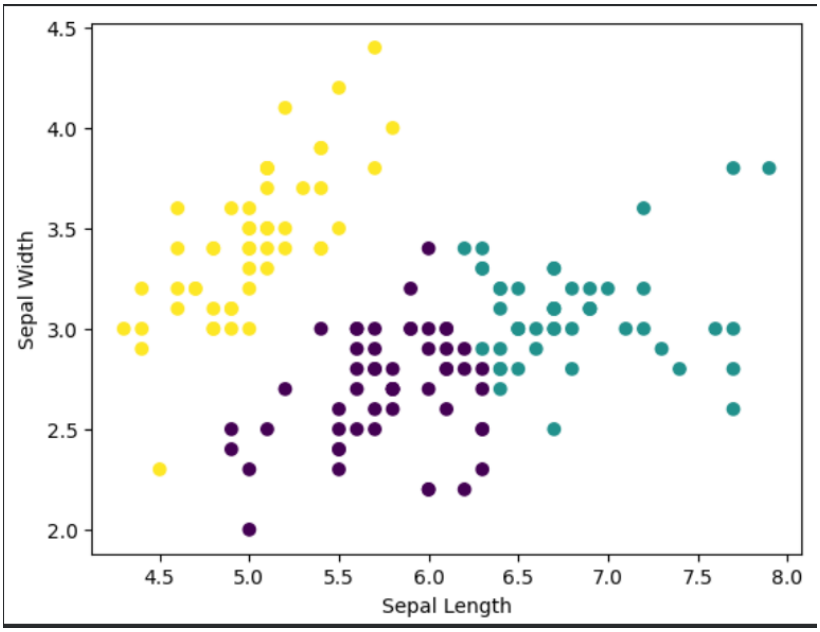
kmeans.fit(X)

plt.scatter(X["sepal_length"], X["sepal_width"], c=kmeans.labels_)
plt.xlabel("Sepal Length")
plt.ylabel("Sepal Width")
plt.show()
```

Dataset Link

- https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv

Output:



CHAPTER 4

SAMPLE PROGRAMS LIST

1. Write a Python program to load a CSV dataset using Pandas and display the first 10 records, last 10 records, and the shape of the dataset.
2. Write a Python program to analyze a CSV dataset by displaying the column names, data types, and summary statistics using Pandas.
3. Write a Python program to visualize a CSV dataset using Matplotlib by plotting a bar chart and a histogram.
4. Write a Python program to find the total number of records, unique values in a column, and frequency count of each category in a CSV dataset using Pandas.

CHAPTER 5

SAMPLE PROGRAMMING EXAMPLES

1. Write a Python program to load a CSV dataset using Pandas and display the first 10 records, last 10 records, and the shape of the dataset.

```
import pandas as pd

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Display first 10 records
print("First 10 Records")
print(data.head(10))

# Display last 10 records
print("\nLast 10 Records")
print(data.tail(10))

# Display shape
print("\nShape of Dataset")
print(data.shape)
```

2. Write a Python program to analyze a CSV dataset by displaying the column names, data types, and summary statistics using Pandas.

```
import pandas as pd

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Display column names
print("Column Names")
print(data.columns)

# Display data types
```

```
print("\nData Types")
print(data.dtypes)

# Display summary statistics
print("\nSummary Statistics")
print(data.describe())
```

3. Write a Python program to visualize a CSV dataset using Matplotlib by plotting a bar chart and a histogram.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Bar Chart
data["species"].value_counts().plot(kind="bar")
plt.title("Species Count")
plt.show()

# Histogram
data["sepal_length"].plot(kind="hist")
plt.title("Sepal Length")
plt.show()
```

4. Write a Python program to find the total number of records, unique values in a column, and frequency count of each category in a CSV dataset using Pandas.

```
import pandas as pd

# Load dataset
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")

# Total records
print("Total Records:", len(data))

# Unique values
print("\nUnique Species")
print(data["species"].unique())

# Frequency count
print("\nFrequency Count")
print(data["species"].value_counts())
```

SAMPLE VIVA QUESTIONS

Experiment 1: Load a CSV Dataset

1. What is Pandas?
2. What is a DataFrame?
3. What is a CSV file?
4. Which function is used to read a CSV file?
5. What is the purpose of head()?
6. What does shape return?
7. What is the difference between rows and columns?
8. Why do we use Pandas?

Experiment 2: Data Preprocessing

1. What is data preprocessing?
2. What are missing values?
3. Which function is used to replace missing values?
4. What is the purpose of fillna()?
5. Why are duplicate records removed?
6. Which function removes duplicate rows?
7. What is the purpose of rename()?
8. Why is data cleaning important?

Experiment 3: Simulating HDFS Operations

1. What is HDFS?
2. What does HDFS stand for?
3. Which library is used to create folders in Python?
4. What is the purpose of the os library?
5. What is the purpose of the shutil library?
6. Which function is used to copy files?
7. Which function is used to move files?
8. Which function is used to delete files?

Experiment 4: Filtering, Sorting, Group By and Aggregation

1. What is filtering?
2. What is sorting?
3. Which function is used for sorting?
4. What is Group By?
5. What is aggregation?
6. Name two aggregation functions.
7. What does mean() calculate?

Experiment 5: JSON CRUD Operations

1. What is JSON?
2. What does CRUD stand for?
3. Why is JSON used?
4. How do you create a JSON object?
5. What is an attribute in JSON?
6. How do you update a JSON record?
7. How do you delete a JSON record?
8. Where is JSON commonly used?

Experiment 6: JSON Search and Filtering

1. What is document search?
2. How do you search a JSON record?
3. What is filtering?
4. What is a key-value pair?
5. How do you access a value in JSON?
6. How is JSON related to MongoDB?
7. Why is JSON suitable for NoSQL databases?
8. What is a document database?

Experiment 7: Word Count using MapReduce

1. What is MapReduce?
2. What is the Map phase?
3. What is the Reduce phase?
4. Why is Word Count a popular MapReduce example?
5. What data structure is used in this program?
6. What does split() do?
7. What is a dictionary in Python?
8. How are repeated words counted?

Experiment 8: Maximum, Minimum, Average and Sum

1. What is aggregation?
2. Which function finds the maximum value?
3. Which function finds the minimum value?
4. Which function calculates the sum?
5. How is the average calculated?
6. What is the purpose of len()?
7. What type of data is stored in the list?
8. Why are aggregation operations important in Big Data?

Experiment 9: Linear Regression

1. What is Machine Learning?
2. What is Linear Regression?
3. What is the dependent variable?
4. What is the independent variable?
5. Which library is used for Linear Regression?

6. What is prediction?
7. Why is a regression line used?

Experiment 10: K-Means Clustering

1. What is clustering?
2. What is K-Means?
3. What does K represent?
4. Which library is used for K-Means?
5. What is a cluster?
6. Is K-Means supervised or unsupervised learning?
7. What is the purpose of fit()?
8. Why is the Iris dataset commonly used?